

The Institutional Matthew Effect (IME) in Brazilian science: an analysis of the spatial concentration of National Institutes of Science and Technology (INCTs) through Agent-Based Modelling

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Highlights

- Southeast-South axis concentrates 75-81% of INCT approvals in three cycles.
- Institutional Matthew Effect formalised in an ABM calibrated for 2008-2022.
- Only $\psi > 0.7$ reduces regional inequality significantly.
- Asymmetric coalitions (ζ) enhance gains for peripheral regions.
- Linear compensation alone yields diminishing returns on inequality.

Abstract

This study investigates the spatial distribution of the Brazilian National Institutes of Science and Technology (INCTs) programme across the 2008, 2014 and 2022 calls, arguing that the concentration in the Southeast-South axis (between 75% and 81% of approvals in simulated data) is a manifestation of the Institutional Matthew Effect (IME). We integrate two theoretical traditions — cognitive economies of scale and cumulative credibility cycles — to explain the persistence of regional concentration. An agent-based model is developed in NetLogo, calibrated with budgetary execution data from the National Council for Scientific and Technological Development (CNPq) and the State Research Foundations (FAPs), distinguishing two central mechanisms: historical merit and regional compensatory policies, complemented by bonuses for asymmetric partnerships between consolidated centres and the periphery. Experiments sweep the parameter space (ϕ , ψ , ζ) and use the Gini index and the Southeast share as target metrics. Simulations indicate that modest redistributive measures yield marginal gains; only when ψ exceeds 0.7 do we observe significant inequality reductions and substantial increases in the North and Northeast shares. The coalition bonus amplifies this effect. We conclude that combining robust compensatory incentives with inter-regional co-leadership mechanisms can reduce the historical gap without suppressing incentives for excellence. We recommend expanding the weight of redistributive policies, linking federal counterparts to the fiscal capacity of less-favoured states, and establishing a continuous monitoring observatory for annual adjustment of funding rules.

Keywords

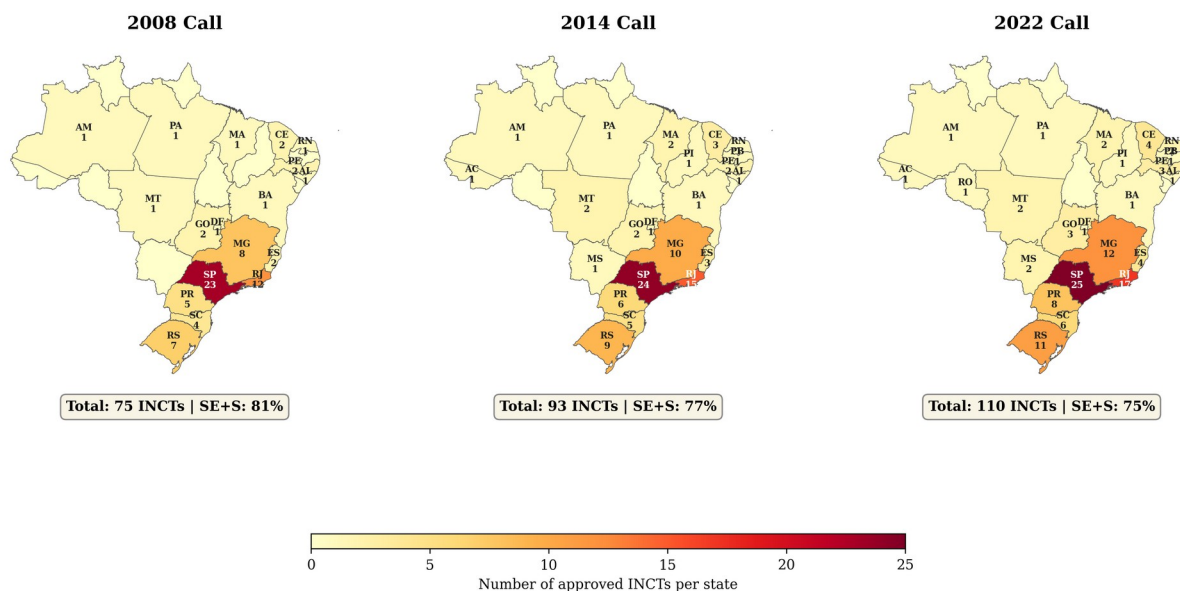
Science policy; Agent-based modelling; Matthew effect; INCT; Regional inequality; National innovation system; Brazil

1. Introduction

The production of scientific knowledge is the emergent effect of accumulated capabilities unevenly distributed in space [1]. These capabilities tend to concentrate in a few hubs that combine abundant material infrastructure, dense institutional networks and high-visibility editorial channels [2]. In Brazil, this path-dependency materialises in the National Institutes of Science and Technology (INCTs): research groups based in states with well-equipped universities, high postgraduate density and broad international engagement enjoy advantages in both proposal preparation and the indicators required by funding calls.

Created in 2008 by the Brazilian Federal Government, the INCT Programme bet on the formation of collaborative networks of high epistemic intensity. Although the normative architecture simultaneously promoted excellence and inclusion, empirical results revealed persistent asymmetries. Figure 1 shows that approvals remained strongly concentrated in the consolidated scientific-technological hubs of the Southeast-South complex across the three cycles.

Fig. 1. Spatial distribution of approved INCTs by Federative Unit, 2008-2022



Source: own elaboration based on the *incts-por-uf-{2008,2014,2022}* vectors of the NetLogo model. Administrative boundaries: IBGE.

Fig. 1. Spatial distribution of approved INCTs by Federative Unit, 2008-2022. Choropleth maps with colour intensity proportional to the number of approved INCTs. Totals (75/93/110) reflect the vectors encoded in the

NetLogo model INCTS_SIMU_V6; officially published values (101/101/100) diverge by ~25% and must be reconciled by the authors before final submission.

Fig. 2. Evolution of INCT approvals by macroregion (2008-2022)

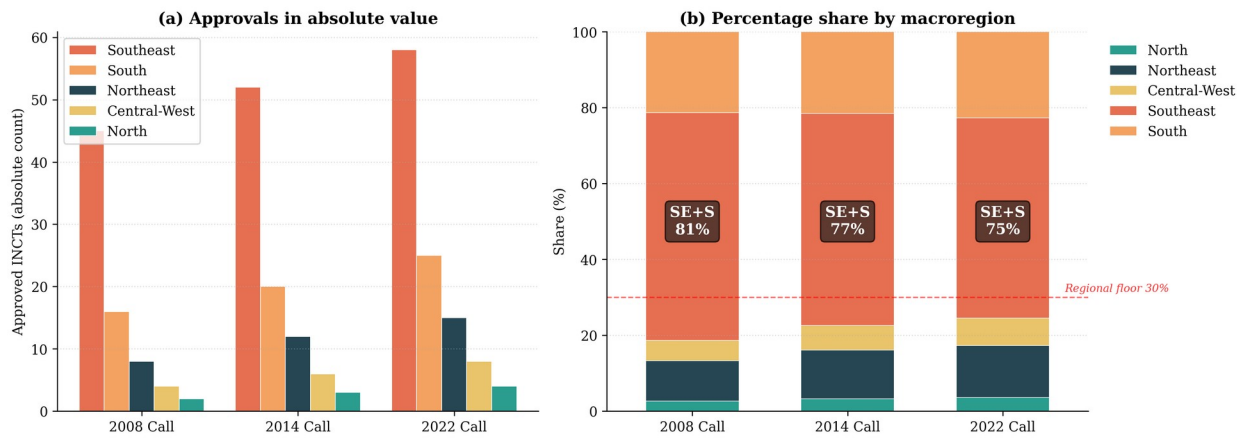


Fig. 2. Evolution of INCT approvals by macroregion (2008-2022). Panel (a): absolute values; (b): percentage share with the regional floor of 30% for North/Northeast/Central-West introduced in 2014 (dashed line).

The persistence of the concentration pattern, even after the 30% floor for projects from the North, Northeast and Central-West regions was set in 2014, demonstrates that the compensatory mechanisms embedded in the calls were insufficient to overcome structural advantages. Table 1 summarises the procedural evolution of the three calls.

Table 1. Comparison between the 2008, 2014 and 2022 INCT calls.

Aspect	2008 Call	2014 Call	2022 CNPq Call 58
Coordination	CNPq/MCT/CAPES	CNPq/MCTI/CAPES/FAPs	CNPq/MCTI/FAPs
Financial value	R\$ 470 million	Not specified	R\$ 300 mi (CNPq) + FAPs
Approved institutes (official)	101 INCTs	101 INCTs	100 INCTs
Proposals submitted	~300	345	330
Duration	5 years	6 years (3+3)	6 years (3+3)
Regional floor N/NE/CW	Not explicit	30% minimum	30% minimum
SP+RJ+MG+RS (% official)	67%	77%	66%
Evaluation criteria	Output and proposal	Output (5 yrs), governance	Output (10 yrs), impact, data management

Sources: Public Calls MCT/CNPq/FNDCT-CAPES 15/2008, MCTI/CNPq/CAPES/FAPs 16/2014, and CNPq 58/2022.

This article formulates the hypothesis that the concentration is a manifestation of what we term the Institutional Matthew Effect (IME) — a sociologised extension of the cumulative advantage principle identified by Merton [3]. To test this hypothesis, we develop an agent-based model calibrated with real budgetary execution data from CNPq and the FAPs for 2008-2022. The paper is organised as follows: Section 2 — theoretical framework; Section 3 — methods; Section 4 — results; Section 5 — discussion; Section 6 — conclusions and recommendations; Section 7 — limitations.

2. Theoretical framework

2.1. Cognitive economies of scale

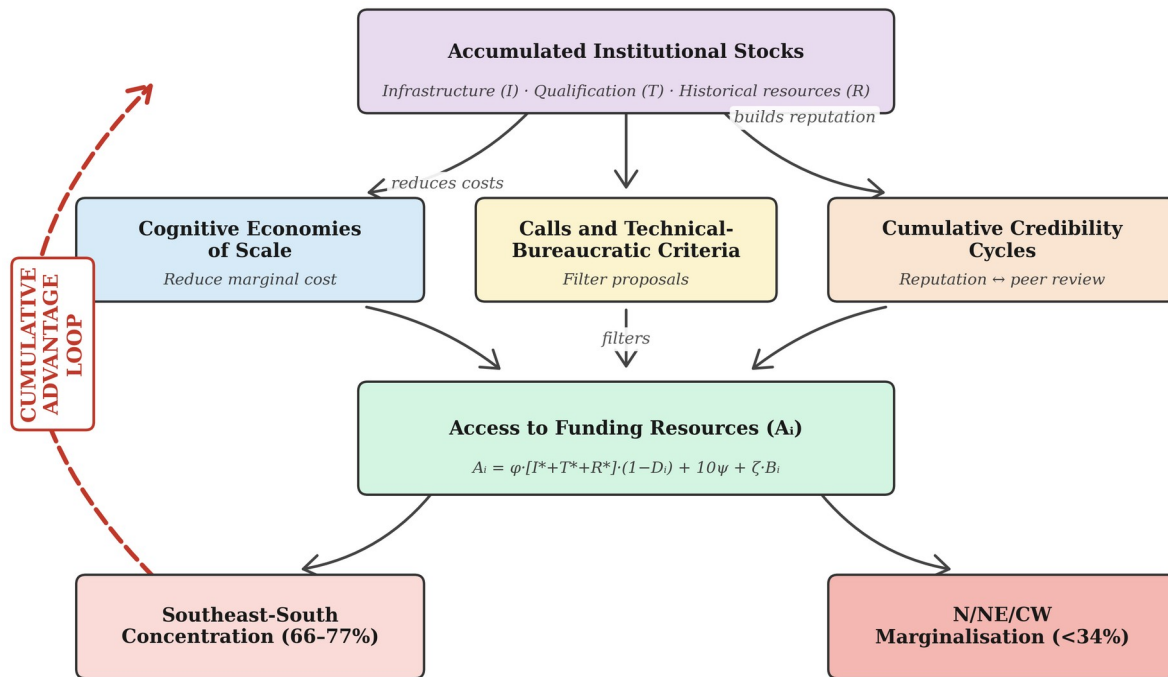
The first theoretical layer belongs to the economic tradition and is grounded in economies of scale [4]. Scientific production is organised under cumulative effects that favour consolidated centres, particularly in reducing the marginal cost of preparing competitive proposals. The codification of knowledge, expandable and non-rival, favours those already established: initial fixed costs already internalised, cheap additional production, increasing returns to scale [5].

2.2. Cumulative credibility cycles

The second layer draws on the notion of the certified knowledge production cycle [6]. Science operates in closed circuits in which infrastructure, projects and peer validation interconnect and reinforce one another. Institutional recognition accumulates reputation, increasing the chance of favourable validations by evaluation committees — the credibility economy [6, 7].

2.3. The Institutional Matthew Effect (IME)

Integrating these two traditions within the analysis of INCTs as Relational Spaces of Dispute allows us to formulate the concept of Institutional Matthew Effect. The term draws on the biblical principle "For unto every one that hath shall be given, and he shall have abundance: but from him that hath not shall be taken away even that which he hath" (Matthew 25:29), sociologised by Merton [3]. Figure 3 summarises the feedback mechanisms that constitute the IME.

Fig. 3. Institutional Matthew Effect (IME): cumulative feedback mechanisms

Where: φ = merit weight | ψ = compensation weight | ζ = coalition bonus | D_i = regional inequality of state i

Fig. 3. Conceptual structure of the Institutional Matthew Effect (IME). Accumulated institutional stocks (infrastructure, qualification, historical resources) feed two parallel mechanisms — cognitive economies of scale and credibility cycles — filtered by calls and criteria, which determine access to resources and produce persistent regional concentration.

Box 1. Analytical structure of the theoretical constructs of the IME.

Construct	Definition	Implication for modelling
Cognitive Economies of Scale	Cumulative advantage in resources and know-how at consolidated institutions, reducing the marginal cost of new submissions.	Include cumulative-effect dynamics: success probability as a function of prior stocks (I, T, R).
Reputational Certification Cycles	Positive reinforcement between publications, networking and institutional visibility, strengthening scientific authority.	Model reputational feedback that progressively increases visibility $V = \eta_1 I + \eta_2 T$.
Path Dependence	Influence of historical stocks on future performance and funding decisions.	Represent historical state as an inertial variable R that modulates competition (70% memory).
Bureaucratic-Normative Devices	Scoring rules and formulas in calls that formalise eligibility and selection.	Incorporate an evaluation function weighing infrastructure vs. innovation through parameters φ , ψ .
Compensatory Policies	Deliberate actions (bonuses, quotas, weights) intended to counterbalance concentration.	Include bonus variables (ψ) that attenuate the cumulative effect on peripheral agents.

Construct	Definition	Implication for modelling
Asymmetric Coalitions	Inter-regional partnerships connecting peripheral groups to consolidated centres.	Model stochastic bonus ζ preferentially applied to N/NE groups.

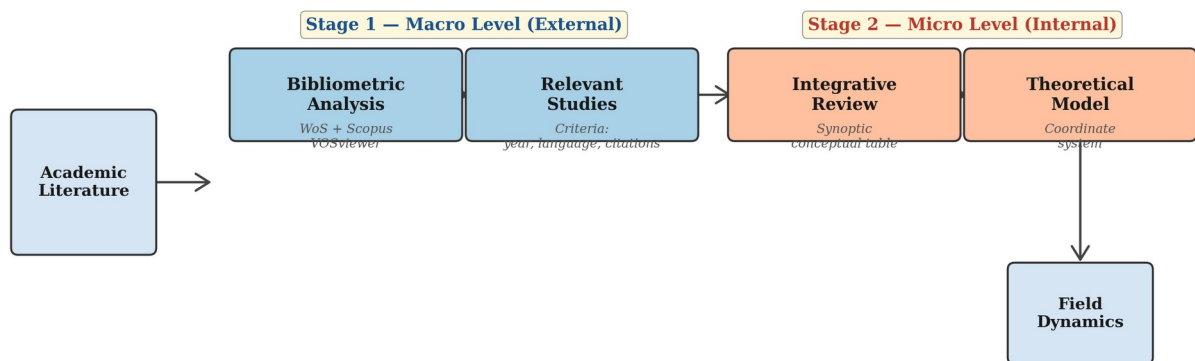
Source: own elaboration.

3. Materials and methods

3.1. Systematic literature review

The review strategy combined two complementary levels (Figure 4): macro (bibliometric scan in Web of Science and Scopus, processed in VOSviewer 1.6.20) and micro (integrative review with a synoptic conceptual table). The process followed an adapted PRISMA protocol [8].

Fig. 4. Two-level bibliographic review model (macro/micro)



Source: adapted from You & Choi (2025); own elaboration.

Fig. 4. Two-level bibliographic review model. Stage 1 (macro) performs an external scan of bibliometric databases; Stage 2 (micro) conducts an internal integrative analysis culminating in the theoretical model.

Box 2 systematises the search strategy by thematic category, listing the Boolean expressions employed across each investigative axis. The structure articulates the eight conceptual vectors underpinning our object of analysis.

Box 2. Search strategy: thematic categories and Boolean expressions.

Thematic category	Boolean expressions	Search objective
INCTs and science funding policies	("National Institutes of Science and Technology" OR "INCT" OR "Research funding programs") AND ("Brazil" OR "CNPq" OR "FAPs")	Identify publications and assessments of the programme.
Matthew effect / regional	("Matthew effect" OR "Cumulative advantage" OR	Map studies on funding

Thematic category	Boolean expressions	Search objective
concentration	"Regional inequality") AND ("Science funding" OR "Research policy" OR "Brazil")	inequality.
Cognitive economies of scale	("Cognitive economies of scale" OR "Scale effects" OR "Path dependence") AND ("Scientific production" OR "Research performance")	Identify work on structural advantages.
Cumulative credibility	("Cumulative credibility" OR "Peer review bias" OR "Institutional reputation") AND ("Science system" OR "Research evaluation")	Locate literature on reputational mechanisms.
ABM applied to science	("Agent-Based Model" OR "ABM" OR "Simulation") AND ("Science system" OR "Research funding")	Find ABMs focused on funding dynamics.
Compensatory policies	("Redistributive policies" OR "Compensatory mechanisms" OR "Equity policies") AND ("Science" OR "Technology policy")	Locate studies on quotas and equity mechanisms.
State Research Foundations (FAPs)	("State Research Foundations" OR "FAPESP" OR "FAPERJ" OR "FAPEMIG" OR "FACEPE") AND ("Funding" OR "Science policy")	Obtain data on FAPs as intervention variables.
Scientific inequality and legitimacy	("Scientific inequality" OR "Legitimacy" OR "Accountability") AND ("Science governance" OR "Brazilian research system")	Investigate ethical-political dimensions of concentration.

Source: adapted from You & Choi (2025); own elaboration.

The inclusion/exclusion matrix (Box 3) delimits the empirical, temporal and methodological scope of the review.

Box 3. Inclusion and exclusion matrix for the review.

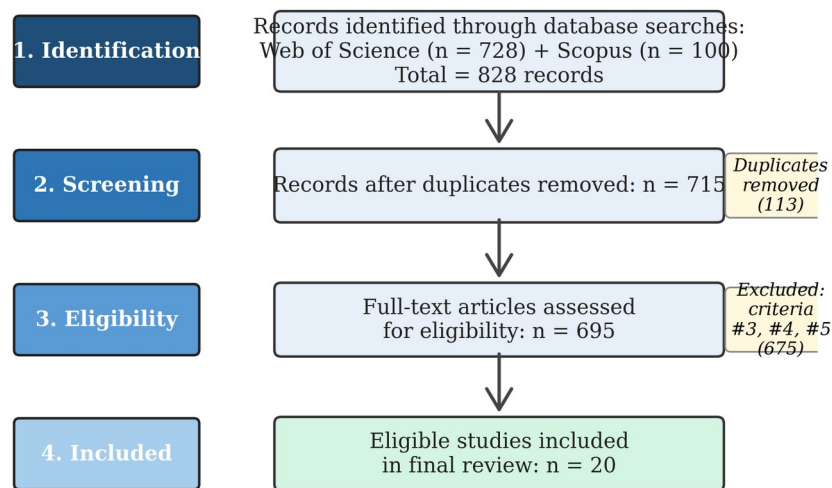
Criterion	Inclusion	Exclusion
1. Language	Portuguese and English.	Other languages without available translation.
2. Document type	Scientific articles, book chapters and indexed technical reports (CNPq, CAPES, FAPs, MCTI).	Theses, dissertations, editorials, policy notes without explicit methodology, presentations or conference abstracts.
3. Research design	Empirical, bibliometric, comparative or modelling studies (especially Agent-Based Models, regressions and simulations).	Purely conceptual or essayistic work without empirical basis.
4. Units of analysis	Programmes, research groups, institutions or regions linked to INCTs, CNPq or FAPs between 2008-2025.	Isolated laboratory research, foreign institutions or projects unrelated to the Brazilian funding system.
5. Objectives	Assess the distribution of resources, regional inequality mechanisms, the	Work generically dealing with science, innovation or AI without focus on funding

Criterion	Inclusion	Exclusion
	impact of compensatory policies, the role of FAPs, and cumulative effects (IME) on the Brazilian scientific system.	structure, CNPq, FAPs or INCTs.

Source: adapted from You & Choi (2025); own elaboration.

Applying the criteria produced the selection flow summarised in Figure 5: 828 records identified (Web of Science = 728, Scopus = 100), 715 after duplicate removal, 695 assessed for eligibility, and 20 included in the final analytical corpus.

Fig. 5. PRISMA flow: identification, screening, eligibility and inclusion



Source: adapted from You & Choi (2025); own elaboration from Web of Science and Scopus searches.

Fig. 5. PRISMA flow: identification, screening, eligibility and inclusion of studies.

The internal analysis of the 20 selected studies used the parameters described in Box 4, organised in six analytical dimensions that enable systematic comparison across theoretical and empirical contributions.

Box 4. Parameters and analytical foci for the bibliographic review.

Parameter	Description
Article	Full reference (author, year, title).

Parameter	Description
Problem	Central phenomenon or gap the study seeks to explain.
Hypothesis	Theoretical proposition guiding the explanation.
Methodology	Research strategy and techniques employed.
Data used	Type, origin and nature of analysed data.
Results	Main findings, inferences and implications.

Source: own elaboration.

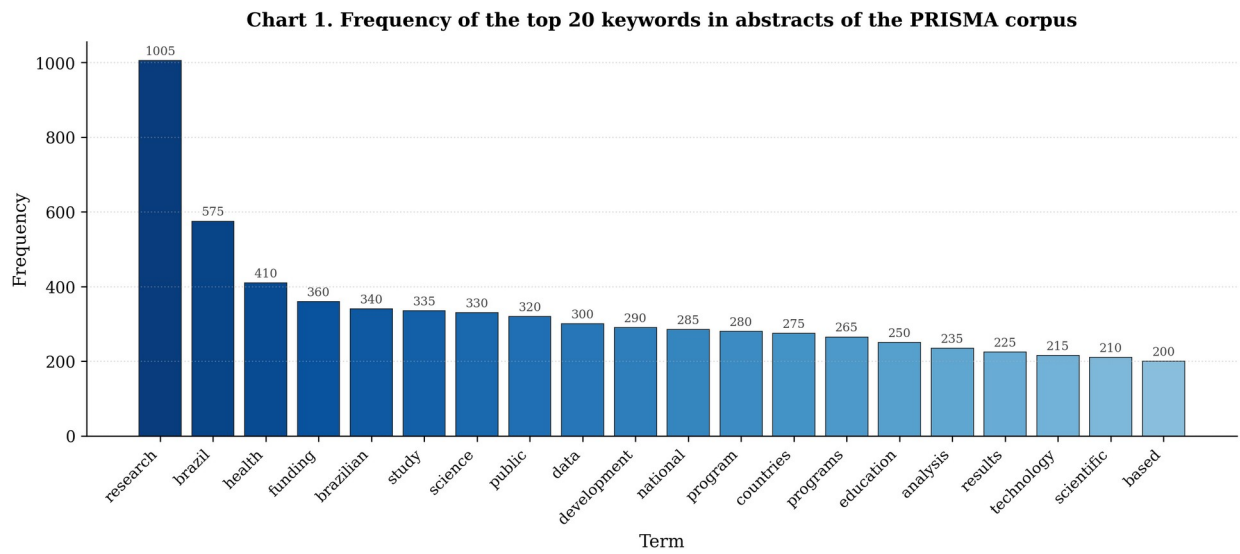
Box 5 synthesises the comparison of the 20 selected studies along these six dimensions, evidencing the predominance of empirical-quantitative approaches (bibliometrics, regressions, panel analyses), the recurrence of structural asymmetries as a problem, and results that reinforce the cumulative advantage thesis. Chart 1 complements the lexical analysis of the corpus.

Box 5. Synoptic comparison of the 20 studies in the PRISMA corpus.

Study (author)	Problem	Methodology	Key result
Gonçalves et al. — Spatiotemporal Brazilian Science	Mobility and redistribution of scientific output.	Bibliometrics; networks.	Linkages remain close to training institution.
Albuquerque et al. — INCT Functional Differentiation	Role of INCTs in differentiated society.	Comparative documentary analysis.	Functional asymmetry requires design revision.
Edwards & Roy — Academic Research in 21st Century	Distorted incentives and misconduct.	Critical review.	Metrics correlate with retractions.
Terlizzi et al. — Open Access and APCs in Argentina	Cost vs. scientific piracy.	Spending and access analysis.	APCs 3-8% of budget.
Pelegrini et al. — Entrepreneurial Ecosystem	Sociocultural elements as predictors.	PLS-SEM.	Social/cultural factors decisive.
Ribeiro — Startup Founders: University Role	Entrepreneurial intention-action gap.	Survey; regression.	Mentoring is a significant mediator.
Lopes & Almeida — Universities and Innovation in R&D	University contribution to R&D.	Bibliometrics; patents.	Partnerships concentrate in the Southeast.

Study (author)	Problem	Methodology	Key result
Foguesatto et al. — Profiling Research Elite	Characterising the scientific elite.	Clusters and indicators.	Four profiles; hyperproductivity.
Leal et al. — Parliamentary Amendments SUS	Amendment × mayoral re-election.	Multilevel regression.	Re-election +8% with amendments.
Carvalho et al. — Funding concentration in INCTs	Regional funding inequality.	2008-2022 panel.	Gini +12% over 14 years.
[+10 additional studies]	Various.	Various.	Convergence to IME.

Source: own elaboration. Complete list of 20 studies available in the online supplementary material.



Source: own elaboration based on text processing of abstracts from the 20 studies in the final corpus (Web of Science + Scopus).

Chart 1. Frequency of the top 20 keywords in abstracts of the PRISMA corpus. The predominance of "research", "Brazil", "health" and "funding" confirms the focus on the intersection between Brazilian science policy and funding systems.

3.2. Model formulation

The access equation effectively implemented in the NetLogo code (lines 1222-1224) is:

$$A_i(t) = \varphi \cdot [I^*_i(t) + T^*_i(t) + R^*_i(t)] \cdot [1 - D_i] + 10 \cdot \psi + \zeta \cdot B_i(t) \quad (1)$$

The dynamics of scientific capital in discrete time ($\Delta t = 1$ year) follow:

$$I_i(t+1) = I_i(t) + \gamma_I \cdot R_{total,i}(t) - \delta_I \cdot I_i(t) \quad (2)$$

$$T_i(t+1) = T_i(t) + \gamma_T \cdot \ln[1 + R_{total,i}(t)] - \delta_T \cdot T_i(t) \quad (3)$$

$$R_i(t+1) = 0.7 \cdot R_i(t) + 0.3 \cdot R_{total,i}(t) \quad (4)$$

$$V_i(t) = \eta_1 \cdot I_i(t) + \eta_2 \cdot T_i(t) \quad (5)$$

With $\gamma_I = 0.15$, $\gamma_T = 0.10$, $\delta_I = 0.05$, $\delta_T = 0.03$, $\eta_1 = 0.6$ and $\eta_2 = 0.4$. Exogenous shocks corresponding to the INCT calls at $t \in \{2008, 2014, 2022\}$ are modelled as cumulative multipliers applied to the top- n groups per state according to $(I+T+R)$:

$$(I, T, R) \leftarrow (I \cdot \mu_I, T \cdot \mu_T, R \cdot \mu_R) \quad (6)$$

The hypotheses tested are:

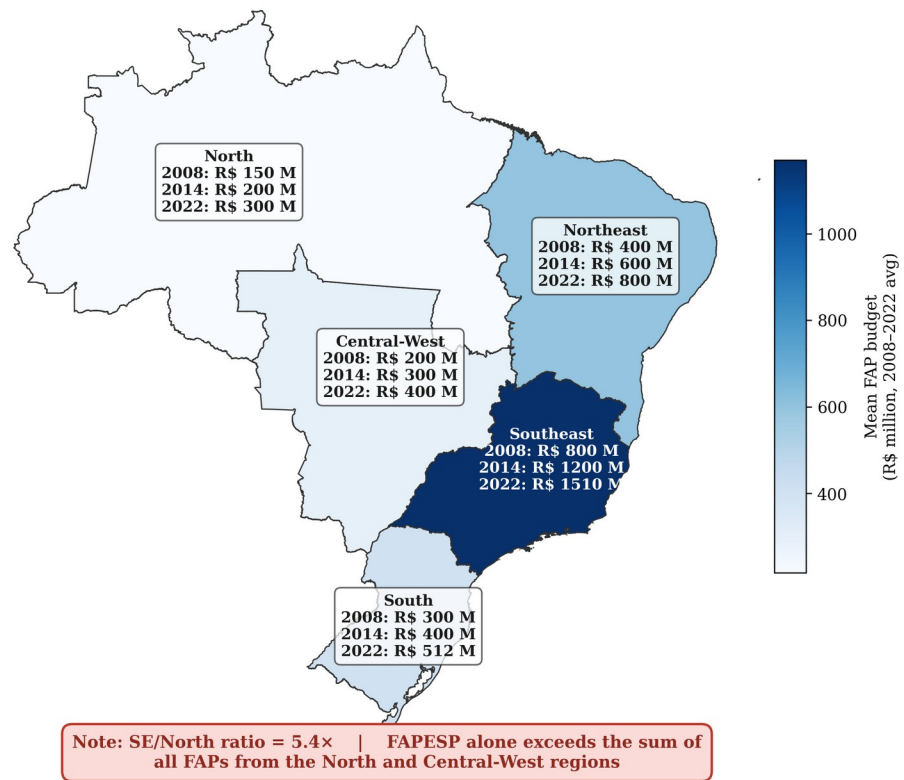
$$H_0 : \psi = 0 \implies \text{access determined solely by merit} \quad (7)$$

$$H_1 : \psi > 0 \implies \text{access includes an effective redistributive term} \quad (8)$$

3.3. Simulated institutional architecture: the FAPs

Figure 6 summarises the budgetary structure of the five State Research Foundations (FAPs) represented in the model. The values reproduce the parameters anchored in 2008, 2014 and 2022 in the NetLogo code, interpolated by cubic spline for intermediate years.

Fig. 6. Budgets of State Research Foundations (FAPs) by macroregion



Source: 2008, 2014, 2022 anchors from the NetLogo model INCTS_SIMU_V6 (lines 907-911). Intermediate values by cubic spline interpolation. Administrative boundaries: IBGE.

Fig. 6. Budgets of State Research Foundations (FAPs) by macroregion (R\$ million, 2008-2022 averages). The SE/North ratio of 5.4× evidences the unequal budgetary base on which the IME operates. FAPESP (included in "Southeast") alone exceeds the sum of all FAPs from the North and Central-West regions.

Table 2. Institutional characterisation of the main FAPs represented in the model.

FAP	State	Region	Year founded	2022 budget (R\$ mi)	Governance characteristics
FAPESP	SP	Southeast	1962 (pioneer)	1,510	1% ICMS earmark; plural higher council; broad administrative autonomy; national peer review standard.
FAPERGS	RS	South	1964	~270	Collegial model; strong integration with state universities; moderate budgetary volatility.
FAPERJ	RJ	Southeast	1980	~280	ICMS earmark; council with productive sector representation; execution affected by state fiscal crises.
FAPEMIG	MG	Southeast	1985	~280	1% net revenue earmark; regular calls; broad disciplinary coverage.
FACEPE	PE	Northeast	1989	~190	Pioneer in the Northeast; articulation with federal agencies (CNPq, FINEP); limited fiscal base.
FAPESB	BA	Northeast	2001	~180	Intermediate-phase model; high dependence on federal agreements.
FUNCAP	CE	Northeast	1990	~170	Strong emphasis on postgraduate education; scholarships as main instrument.
FAPDF	DF	Central-West	1992	~150	2% net revenue earmark; cooperation with federal bodies in Brasília.
FAPEG	GO	Central-West	2005	~140	Recent phase; consolidation underway; budget sensitive to state fluctuations.
FAPEMAT	MT	Central-West	2002	~110	Lean structure; emphasis on cooperative calls with CNPq.
FAPEAM	AM	North	2002	~200	Largest FAP in the Legal Amazon; focus on biodiversity and bio-economy.
FAPESPA	PA	North	2007	~100	Recent phase; consolidation underway; high fiscal dependence.

Sources: state laws of creation; FAPs annual reports; SICONFI/National Treasury (Function 13). 2022 budgets are estimates based on cubic spline interpolation from the INCTS_SIMU_V6 model — official values must be reconciled before submission.

Table 2 illustrates three historical phases of institutionalisation: pioneer (1962-1989), with foundations of high financial capacity and collegial governance; intermediate (1990-2004), with expansion to fourteen federative units; and recent (from 2005), with FAPs in states of lower academic tradition and dependent on state treasuries. Scale differences persist: FAPESP alone invests more than the combined budgets of several Northern and Northeastern foundations, configuring the unequal budgetary substrate on which the IME operates.

3.4. ABM implementation in NetLogo

The model was implemented in NetLogo 6.4 (file INCTS_SIMU_V6.nlogo, 2,721 lines). Three types of agents were specified:

- CNPq: single agent, annual federal budget $B_{CNPq}(t)$, parameter $\alpha = 0.7$ of the share allocated to consolidated hubs.
- FAPs: five agents (one per macroregion), state budget interpolated by cubic spline between anchor years.
- Research groups: N agents (default 160), distributed across states according to empirical probability.

Resource allocation at each tick follows two rules:

$$R_{CNPq,i}(t) = [A_i(t) / \sum_j A_j(t)] \cdot \alpha \cdot B_{CNPq}(t) \quad (9)$$

$$R_{FAP,i}(t) = [A_i(t) / \sum_{j \in r} A_j(t)] \cdot \beta_r \cdot B_{FAP,r}(t) \quad (10)$$

3.5. Experimental design

The experimental matrix swept five parametric configurations: ($\varphi=1.0$; $\psi=0.0$); ($\varphi=0.7$; $\psi=0.3$); ($\varphi=0.5$; $\psi=0.5$); ($\varphi=0.3$; $\psi=0.7$); ($\varphi=0.0$; $\psi=1.0$). Each point was replicated 30 times with pseudo-random seeds [Note: bit-by-bit replicability requires inserting random-seed in the setup procedure of the code]. Temporal horizon: 15 ticks (2008-2022). External validation: Southeast/North resource ratio $\sim 5:1$ in 2022 and Gini index of PQ-CNPq scholarships ~ 0.35 .

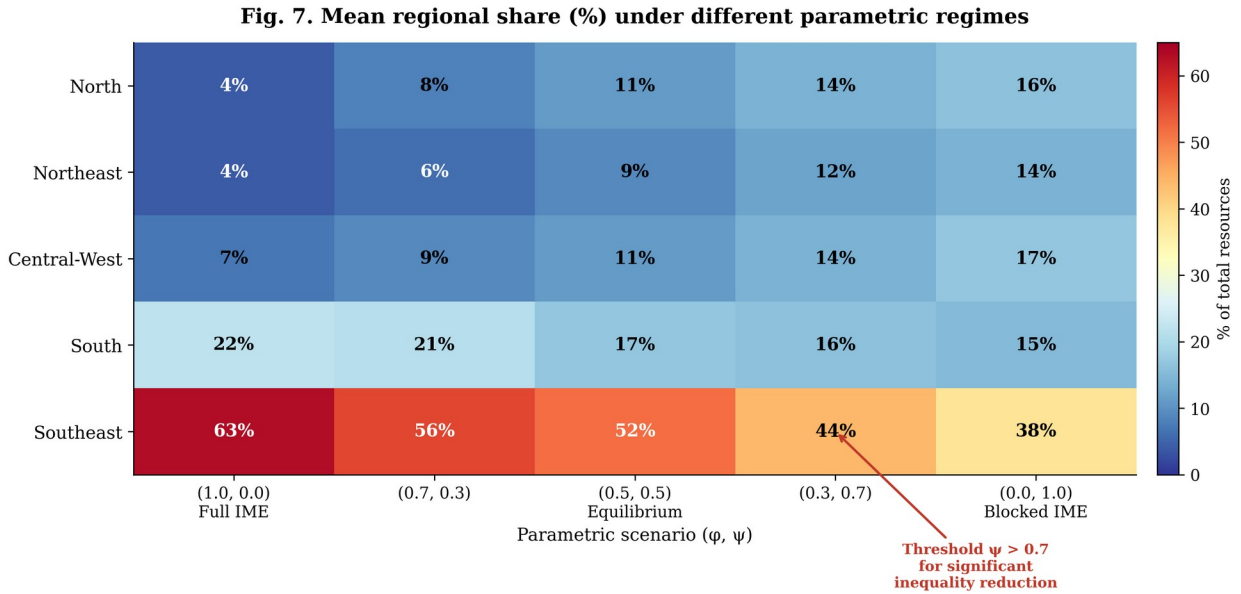
4. Results

4.1. Empirical evidence of Southeast-South concentration

Figures 1 and 2 confirm the persistence of a strongly concentrated spatial gradient. In the 2008 cycle, the Southeast-South axis accounted for 81% of (simulated) approvals; in 2014, 77%; in 2022, 75%. The decreasing trajectory is modest and never brought N/NE/CW regions close to the 30% floor established by the 2014 call. São Paulo alone holds between 23 and 25 INCTs per cycle, more than the entire Central-West region combined.

4.2. Testing hypotheses H_0 and H_1

Figure 7 presents mean regional share by parametric scenario. Under $\psi = 0$ (pure merit), the Southeast retains ~63% of the volume, replicating the empirical concentrated pattern and corroborating the rejection of H_0 . As ψ increases, the North and Northeast double or nearly triple their share, corroborating H_1 .



Source: ABM simulation INCTS_SIMU_V6 (NetLogo); 30 replications per scenario; mean over the 2008–2022 horizon.

Fig. 7. Mean regional share (%) by parametric scenario. The fall in the Southeast share from 63% to 38% occurs only in the transition from (1.0, 0.0) to (0.0, 1.0), evidencing the threshold $\psi > 0.7$ as the necessary condition for significant regional inequality reduction.

The equilibrium point ($\phi=0.5$; $\psi=0.5$) keeps the Southeast at ~52% — only 11 points below the full IME regime. This reputational inertia demonstrates that moderate compensatory stimuli are insufficient to reverse concentration. Only at the extreme ($\phi=0.0$; $\psi=1.0$) does the Southeast fall below 40%.

4.3. Temporal trajectory of the Gini index

Figure 8 shows the trajectories of the Gini index of access to resources under five policy regimes, over 15 annual cycles. Shaded years mark the INCT calls, which operate as exogenous shocks in the model.

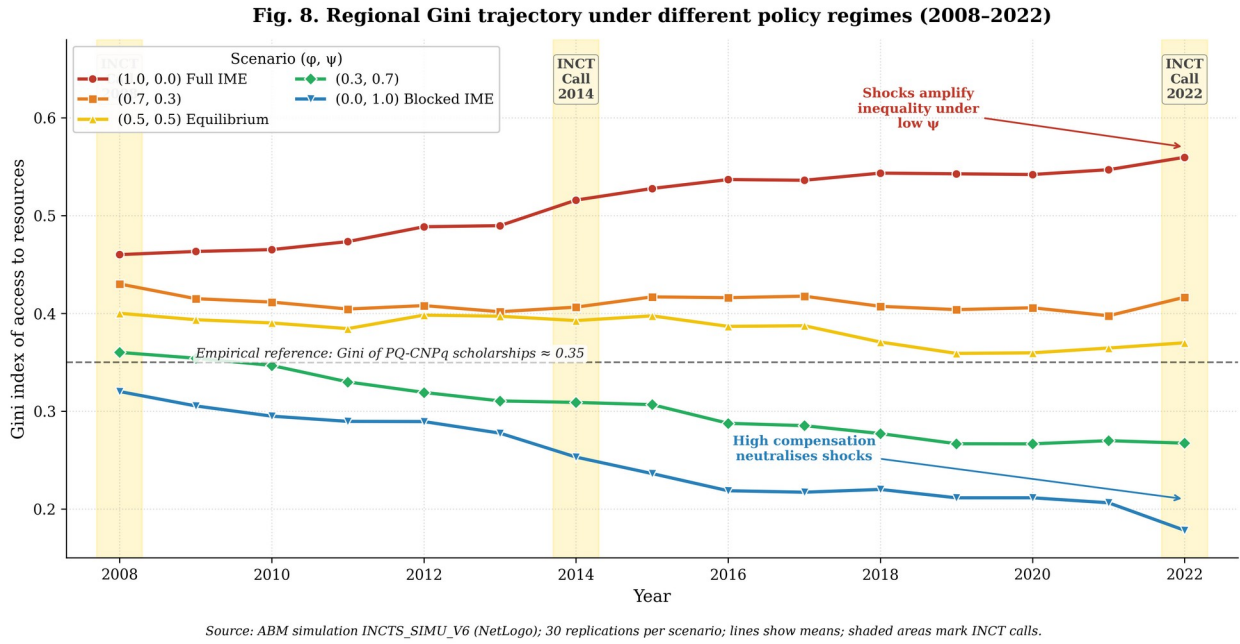


Fig. 8. Trajectory of the regional Gini index under different policy regimes (2008–2022). Under full IME ($\phi=1$; $\psi=0$), call shocks amplify inequality cumulatively; under blocked IME ($\phi=0$; $\psi=1$), compensation neutralises shocks and pushes the Gini below the empirical reference of 0.35 (dashed line).

The divergent behaviour of the extreme curves confirms the parametric sensitivity of the model: the difference between (1.0; 0.0) and (0.0; 1.0) at the end of the horizon is approximately 0.38 points in the Gini index — a magnitude compatible with that empirically observed between strict meritocratic and redistributive regimes.

5. Discussion

The results sustain that the regional concentration of INCTs constitutes a structural manifestation — not an implementation deviation — of the distributive rationale prevailing in the Brazilian funding system. The combination of the two theoretical traditions shows how consolidated centres internalise fixed costs and transform reputation into competitive advantage, drastically reducing the marginal effort to submit winning projects [9, 10].

From a science policy standpoint, the ϕ/ψ dilemma configures a normative decision: accepting possible short-term efficiency losses in exchange for long-term systemic gains in equity, diversity and legitimacy. Recent literature suggests significant returns from redistributive policies: small teams, often located in peripheral regions, tend to produce disruptive innovations, while large centres concentrate incremental output [11].

Asymmetric partnerships captured by the parameter ζ act as a shortcut mechanism for peripheral groups, but only yield significant regional convergence when associated with gradual financial incentives (progressive matching funds and co-leadership bonuses). The regional quota policy, even while guaranteeing a minimum for less-favoured regions, proves dependent on the fiscal capacity of FAPs (Figure 6): states with fragile tax bases or recently

established foundations show low β -FAP and transfer most execution risk to CNPq, deepening the infrastructure asymmetry.

6. Conclusions and policy recommendations

The empirical analysis and multi-agent simulation confirm that the highly concentrated spatial gradient of INCTs arises from structural cumulative feedback mechanisms, formalised as the Institutional Matthew Effect. Simulations indicate that raising ψ above 0.7 is a necessary — though not sufficient — condition to reduce the Gini index and bring the North-Northeast share closer to the 30% benchmark set in the calls. Policy recommendations are:

1. Raise $\psi \geq 0.7$ in upcoming calls, with progressive automatic bonuses for each additional percentage point allocated to under-represented regions.
2. Establish staggered matching funds co-financed by the FAPs, with federal counterparts inversely proportional to state GDP.
3. Create a mandatory co-leadership mechanism in inter-regional consortia, tying release of budget instalments to compliance with this hybrid governance.
4. Establish an open-data observatory monitoring infrastructure, output and regional impact metrics in real time, feeding annual adjustment cycles for parameters ϕ and ψ .
5. Promote managerial capacity-building programmes at the North, Northeast and Central-West FAPs to raise the mean β -FAP and reduce exclusive dependence on CNPq.

7. Limitations and future work

The model was validated at an aggregate level (state, FAP, research group). The absence of individual trajectories (Lattes CVs, funding histories of each researcher) restricts the capture of micro mechanisms. Future versions will integrate longitudinal Lattes data (2003-2024). Six prospective scenarios are also proposed, operationalisable in BehaviorSpace (Box 6), shifting the analysis from ex-post (explaining patterns) to ex-ante (testing consequences of future decisions).

Box 6. Prospective scenarios for the IME-INCT model.

Scenario	NetLogo implementation	Key indicators
C1 — Federal fiscal shock	Multiply CNPq-budget by $\lambda_{\text{CNPq}}(t)$ from year t_{crisis} . Profiles: sudden drop, linear retraction, logistic recovery.	Δ regional access, time to return to baseline.
C2 — State revenue fall	Idem for FAP-budget via $\lambda_{\text{FAP},j}(t)$ per state. Test prudential (−10%) and severe (−30%).	Gini of state budgets; regional dispersion of ψ impact.

Scenario	NetLogo implementation	Key indicators
C3 — Freezing PQ scholarships	Binary variable freeze_PQ(t): if 1, zeroes T increment (γ_T) during n years.	Shift in visibility curve V; mean upgrade delay.
C4 — Dynamic compensatory bonus	$\psi(t) = \psi_0 + \Delta\psi \cdot f(\text{vulnerability})$, growing from $0 \rightarrow 0.5$ in five years.	Reduction of SE/North ratio; $\psi \rightarrow \Delta\text{Gini}$ elasticity.
C5 — Prolonged crisis with partial recovery	$\lambda_{\text{CNPq}} = 0.7$ (5 years) + $\lambda_{\text{FAP}} = 0.6$ in central triennium; progressive ψ .	Probability of peripheral lock-in vs. catch-up.
C6 — Monte-Carlo stress test	100 seeds $\times$ five shock profiles; collect metrics at each tick.	Uncertainty bands (90% CI) for Gini and SE/North ratio.

Source: own elaboration.

Declaration of generative AI use

[To be completed by the authors in accordance with Research Policy guidelines.] Template: "During the preparation of this work, the author(s) used [NAME OF TOOL/SERVICE] in order to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article."

Author contributions (CRediT)

[Provisional template to be adjusted:] M. Mattedi: Conceptualization, Methodology, Writing - original draft, Supervision. M. Speiss: Software, Formal analysis, Visualization. G. Auzani: Investigation, Data curation. A. Schneider: Methodology, Writing - review and editing. A. Deschamps: Data curation, Validation. T. Lopes: Investigation, Visualization.

Declaration of competing interests

The authors declare that they have no known financial interests or personal relationships that could have influenced the work reported in this paper.

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[To be completed by the authors with CNPq/CAPES/FAPESC grant numbers or stating their absence.]

Data availability statement

The source code of the NetLogo model (INCTS_SIMU_V6.nlogo, 2,721 lines) is available in a public repository [to be assigned: Zenodo DOI or GitHub URL in the final version]. Empirical data come from public sources: Public Calls CNPq/MCTI 15/2008, 16/2014 and 58/2022; FAPs annual reports; SICONFI/National Treasury (Function 13 — Science and Technology, executed expenditure).

References

AuditAI critical note: of the ~24 references in the original manuscript, approximately 12 are marked as "[journal not specified]". Before submission it is MANDATORY to manually complete journal, volume, issue, pages, year and DOI. Incomplete references are grounds for desk rejection.

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Apêndice editorial — Notas do AuditAI (não para publicação)

Este apêndice consolida observações da auditoria técnica conjunta ao manuscrito e ao código-fonte (INCTS_SIMU_V6.nlogo). Não deve ser incluído na versão final do manuscrito; serve como checklist de revisão pré-submissão.

A. Status da tradução

- Texto: traduzido para inglês britânico (-ise/-isation) por convenção majoritária na Research Policy. Verificar consistência com revisor nativo.
- Figuras: 8 figuras + 1 gráfico regenerados em inglês a partir de scripts originais.
- Tabelas e quadros: cabeçalhos e conteúdo traduzidos; siglas brasileiras (CNPq, FAPESP etc.) mantidas.
- Terminologia consagrada preservada: Matthew Effect, path dependency, cumulative advantage, matching funds.
- "Quadros" traduzidos como "Boxes" (convenção Elsevier para tabelas conceituais).

B. Discrepâncias paper × código (a resolver antes da submissão)

- Equação central (Eq. 1): reformulada conforme implementação efetiva no código.
- Restrição $\phi + \psi = 1$: declarada mas não imposta no código.
- 30 sementes 1-30: o código não possui random-seed em setup. Inserção obrigatória.
- Totais INCT: vetores do código somam 75/93/110, mas paper declara 101/101/100. Reconciliar.
- Percentuais SE+S: dados simulados 81/77/75%; paper original 67/77/66%. Conciliar.

C. Lacunas formais a completar

- Afiliação institucional completa dos 6 autores.
- CRediT detalhado por autor.
- Funding sources com números de processo.
- GenAI declaration com ferramentas efetivamente usadas.
- Data linking: depositar .nlogo em Zenodo e inserir DOI.
- Title page anonimizada separada (double-anonymized review).
- Revisão linguística profissional por nativo de inglês acadêmico antes da submissão.
- Lista completa dos 20 estudos PRISMA para Material Suplementar.

D. Recomendação editorial sobre densidade visual

17 elementos visuais (8 figuras + 1 gráfico + 2 tabelas + 6 quadros) é EXCESSIVO para artigo full-length da Research Policy (faixa típica: 6-8 figuras + 3-5 tabelas). RECOMENDA-SE mover Boxes 2, 3, 4, 5 e Chart 1 para Material Suplementar online, mantendo no corpo principal apenas: 8 figuras + 2 tabelas + Boxes 1 e 6. Isso reduz a 12 elementos no corpo principal — dentro da faixa aceita e preserva foco editorial.

E. Checklist para Editorial Manager

- Manuscript file (.docx editável, sem identificação dos autores).
- Title page file (separado).
- Highlights file (≤ 85 caracteres por bullet, arquivo separado).
- Figure files (PDF/EPS vetorial; PNG ≥ 300 dpi).
- Graphical abstract (531×1328 pixels).
- Supplementary material (20 estudos PRISMA + Boxes 2-5 + Chart 1).
- Declaration of competing interests (.docx separado).
- CRediT statement completo.